그림 8. 첫 번째 가상 창문에서 PT를 통해 얻어진 온도, 화실에서 2.1m 위.

내용 요약

- 대부분은 표준 편차 내에 있습니다.
- 불확실성을 약 3%로 가지며, 그림 8에서 PT를 통해 추정된 외벽의 노출도를 보면 시뮬레이션에서의 온도가 실험에서의 온도보다 빨리 낮지만, 나무와 폴리에틸렌 레진 외벽 시스템에서 HRR(Heat Release Rate)에 큰 기여를 하여 측정된 온도가 증가하는 것을 볼 수 있습니다.

구체적인 설명

1. 불확실성

- 대부분의 결과는 표준 편차 내에 있습니다.
- 불확실성을 약 3%로 가지고 있습니다.

2. 외벽 시스템

- 그림 8에서 PT를 통해 추정된 외벽의 노출도를 보면 시뮬레이션에서의 온도가 실험에서의 온도보다 빨리 낮습니다.
- 그러나 나무와 폴리에틸렌 레진 외벽 시스템에서는 HRR(Heat Release Rate)에 큰 기여를 하여 측정된 온도가 증가하는 것을 볼 수 있습니다.

참고 문헌 및 기타 설명

- 한 가지 테스트가 다른 테스트와 다르게 나타나는데, 이는 매우 두꺼운 벽돌로 된 외벽 시스템입니다.
- 측정 결과에 따르면, 연소실에서의 판형 온도계 온도가 다른 테스트보다 높고, 연소실에서 2.1m 위의 온도는 낮습니다.
- 이는 외벽 시스템의 두께로 설명할 수 있습니다.
- 방사열은 불꽃의 두께에 따라 달라집니다.
- 외벽의 두께가 증가하면 초기 화실에서 외벽 표면까지의 거리도 증가하여, 실제 연소가 일어나는 위치가 변화하며 판형 온도계의 측정에 영향을 미칩니다.
- 이는 판형 온도계가 작은 참조 표본에서의 효과적인 표면 온도를 측정한다는 사실을 강조합니다.
- 이 효과적인 온도는 평균적 물체에 대한 열 노출의 약한 측정으로 더 대표적입니다.
- 또한, 열 유량 측정에 비해 불확실성은 빨리 작습니다.
- 외벽의 두께는 연기의 이동 거리, 외부와의 방사 열 교환에 영향을 미치며, 일부 열은 하단 가장자리에 흡수됩니다.
- 즉, 아래를 향한 가장자리로 연료실의 연장으로 볼 수 있습니다.
- 시료가 더 두꺼울수록 이러한 요인들이 역학에 더 큰 영향을 미치고 하단 가장자리에 흡수되는 열도 더 많아집니다.

그림 9. 일반적인 외벽(100mm)과 더 두꺼운 외벽(330mm) 시스템에 대한 시뮬레이션된 온도

- 일반적인 외벽(100mm)과 더 두꺼운 외벽(330mm) 시스템에 대한 시뮬레이션된 온도를 보여줍니다.
- 이는 참조 [7]에서 수행된 실험 비교와 같은 질적 행위를 보여줍니다.
- 외벽 앞쪽의 온도가 증가하는 반면, 화실에서 2.1m 위의 외벽 온도는 감소하는 것을 관찰할 수 있습니다.
- 주목할 점은 시뮬레이션에서 기본 화학 및 탄화물 생성을 변경함으로써, 화재 테스트에서 사용된 다양한 연료의 효과를 시뮬레이션할 수 있다는 것입니다.
- 우리는 핵산(표준 연료), 나무 및 프로판을 사용한 시뮬레이션을 수행했습니다.
- 결과는 PT(화실에서 0.5m 떨어진 외벽 앞쪽)와 PT(화실에서 2.1m 떨어진 외벽)에서 측정된 온도에 대한 다양한 연료 간의 차이가 매우 작음을 보여줍니다.

이미지 정보 (Optional)

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